An orthogonal frame for the matter chord module

Moving-core symmetry limits and a positive null candidate

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Abstract

We give an orthogonal realization of the established one-matter chord representation, separating a left q-oscillator from its vacuum multiplicity. An ordinary displacement on the first factor then defines an explicit moving unitary frame on the entire module. In a joint large-displacement, small-deformation limit, every fixed finite word of the Lin–Stanford generators and the Hamiltonian boost converges on the finite-support core to the positive discrete series of $\mathfrak{sl}_2$. This extends the preceding endpoint single-vector comparison to descendants and commutator norms. The combination $E - \widehat{P}$ has a positive, boost-covariant limit with the standard Laguerre spectral measure. Finally, an operator-norm bound on the neutral field controls a specified insertion of the full selected density without replacing it by an energy-only density. The limit retains a null-generator candidate but does not determine a mass or a multiparticle law.

1 Setting and existing constructions

Work after the established fixed-λ, half-filled double scaling, with $q = e^{-\epsilon}$, $\epsilon = 4\lambda$, $K = H/(2J_s) = h/\sqrt{\epsilon}$ and $\hbar = 1$. The one-insertion module has weight $d > 0$, crossing $r = q^d$, and seed $\Omega_d = |d; 0, 0\rangle$. The actual charged curvature channel has $d = 2$. This module is the existing tracial one, not a new independent-probe model.

Lin–Stanford (LS) supply its ladder algebra and length-preserving generators [1]. Okuyama's doubled-space map and Xu's isometric formulation explicitly involve the matter-dependent inner product [2, 3]. A linear tensor map in those coordinates cannot simply be treated as a product-inner-product unitary. We instead orthogonalize using the kernel of a_L. This is an elementary realization of the same representation; neither the factorization idea nor the resulting discrete series is claimed as a new representation-theoretic discovery.

We retain the earlier conventions $J_{ij} = a_i^\dagger a_j - [\bar{n}]$, $c = q^{\bar{n}/2}$, $E = -(J_{LL} + J_{RR})/(2c)$, $B = (J_{LL} - J_{RR})/(2c)$, $P = i(J_{LR} - J_{RL})/2$, and $\widehat{B} = B/\sqrt{1 - c^2}$, $\widehat{P} = P/\sqrt{1 - c^2}$. The rescaling acts separately on each total level. The physical comparison is $B_{\mathrm{tr}} = (K_L - K_R)/2$.

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2 An exact orthonormal tensor frame

Write $[n] = (1-q^n)/(1-q)$ and $(r^2;q)_k = \prod_{j=0}^{k-1}(1-r^2q^j)$. Define the normalized left-vacuum vectors

$$\xi_k = \frac{\prod_{j=0}^{k-1}(a_R^\dagger - rq^j a_L^\dagger)\Omega_d}{\sqrt{[k]!(r^2;q)_k}}, \qquad e_{n,k} = \frac{(a_L^\dagger)^n}{\sqrt{[n]!}}\xi_k. \tag{1}$$

All factors in the numerator commute. Empty products are one.

Proposition 1. *The $e_{n,k}$, $n,k \geq 0$, form an orthonormal basis of the one-matter module. The unitary $\mathcal{F}_{q,d} : \ell^2(\mathbb{N}_0) \otimes \ell^2(\mathbb{N}_0) \to \mathcal{H}_d$, defined by $|n\rangle \otimes |k\rangle \mapsto e_{n,k}$, gives*

$$a_L = a_q \otimes 1, \qquad a_R = a_q \otimes R + 1 \otimes b_q, \qquad \bar{n} = N + K_0 + d, \qquad R = q^{d+K_0}, \tag{2}$$

where tensor symbols are suppressed on the right in $\bar{n}$, and

$$a_q|n\rangle = \sqrt{[n]}|n-1\rangle, \quad b_q|k\rangle = \sqrt{[k](1-r^2q^{k-1})}|k-1\rangle, \quad K_0|k\rangle = k|k\rangle.$$

Proof. Suppose ξ_k is a normalized a_L-vacuum at total level k. The relations $[a_L, a_R^\dagger] = q^{\bar{n}}$ and $a_L a_L^\dagger - q a_L^\dagger a_L = 1$ show that $(a_R^\dagger - rq^k a_L^\dagger)\xi_k$ is again a left vacuum. The q-oscillator relation gives the ladder norm $[n]!$. Different n at the same total level have distinct eigenvalues $[n]$ of the self-adjoint $a_L^\dagger a_L$, hence are orthogonal. Inductively there is exactly one new left-vacuum direction per level, because the level-s space has dimension $s+1$.

Let s_k be the norm of the new vacuum vector. Since a_R commutes with a_L, it maps the vacuum line at level k to the previous one; adjointness gives $a_R\xi_k = s_{k-1}\xi_{k-1}$. The right q-oscillator relation yields

$$s_k^2 = 1 + qs_{k-1}^2 - r^2q^{2k} = [k+1](1-r^2q^k), \qquad s_{-1} = 0.$$

This proves the normalization in (1) and positivity of each new direction. Commuting $a_R^\dagger$ through $(a_L^\dagger)^n$ gives (2); the annihilators follow by adjointness. Each total level now has a complete orthonormal set of $s+1$ vectors. □

For $k = 0$, $e_{n,0}$ is the preceding endpoint basis; its orthogonal f_n is exactly $e_{n-1,1}$. Diagonalizing $a_q + a_q^\dagger$ makes the right Hamiltonian an Al-Salam–Chihara Jacobi operator with parameters $re^{\pm i\theta_L}$, consistent with the known conditional matter measure [3, 5].

3 The moving frame and its limiting generators

Let a_1 be the ordinary oscillator on the first factor, and set $D(-t) = \exp[-t(a_1^\dagger - a_1)]$, $t = \sqrt{m}$. This is an ordinary unitary displacement, not q-thermal evolution. Define

$$\mathcal{U}_{q,d,m} = \mathcal{F}_{q,d}(D(-\sqrt{m}) \otimes 1), \qquad \mathcal{D} = \mathrm{span}\{|j\rangle \otimes |k\rangle : j, k < \infty\}. \tag{3}$$

Its ground vector is the moving coherent state used to compare with the true heat vector in the preceding note.

On the second factor introduce the standard discrete-series operators

$$L_-|k\rangle = \sqrt{k(2d+k-1)}|k-1\rangle, \qquad L_+ = L_-^\dagger, \tag{4}$$

$$E_0 = d + K_0, \qquad B_0 = (L_+ + L_-)/2, \qquad P_0 = i(L_+ - L_-)/2. \tag{5}$$

They act as the identity on the first factor. Their commutators are $[B_0, P_0] = iE_0$, $[E_0, P_0] = iB_0$, $[B_0, E_0] = iP_0$, with $E_0^2 - B_0^2 - P_0^2 = d(d-1)$.

Proposition 2. *Take $m \to \infty$, $\epsilon \to 0$ with $\epsilon m \to 0$ and fixed $d > 0$. For every fixed finite noncommutative polynomial p and every $v \in \mathcal{D}$,*

$$\|p(E, \widehat{B}, \widehat{P}, B_{\mathrm{tr}})\mathcal{U}v - \mathcal{U}p(E_0, B_0, P_0, -B_0)v\| \longrightarrow 0. \tag{6}$$

In particular the three $\mathfrak{sl}_2$ defects and the Casimir defect converge in norm on this moving core.

Here are the coefficients used to prove it. Put

$$R_k = q^{d+k}, \qquad c_{n,k} = q^{(n+k+d)/2}, \qquad s_k = \sqrt{[k+1](1 - r^2 q^k)}, \qquad s_{-1} = 0.$$

The diagonal coefficients are

$$E_{n,k;n,k} = \frac{[d+k] - (1-R_k)^2[n]/2 - s_{k-1}^2/2}{c_{n,k}}, \tag{7}$$

$$\widehat{B}_{n,k;n,k} = \frac{(1-R_k^2)[n] - s_{k-1}^2}{2c_{n,k}\sqrt{1-c_{n,k}^2}}, \qquad \widehat{P}_{n,k;n,k} = 0. \tag{8}$$

The edge $(n,k) \to (n-1, k+1)$ has coefficients

$$-\frac{R_k s_k \sqrt{[n]}}{2c_{n,k}}, \qquad -\frac{R_k s_k \sqrt{[n]}}{2c_{n,k}\sqrt{1-c_{n,k}^2}}, \qquad -\frac{i s_k \sqrt{[n]}}{2\sqrt{1-c_{n,k}^2}} \tag{9}$$

for $E, \widehat{B}, \widehat{P}$, respectively; reverse edges are conjugates. Finally the exact Hamiltonian boost is

$$B_{\mathrm{tr}} = \frac{(a_q + a_q^\dagger) \otimes (1-R) - 1 \otimes (b_q + b_q^\dagger)}{2\sqrt{\epsilon}}. \tag{10}$$

4 Proof of the finite-word limit and operator consequence

Write $\zeta_j = D(-\sqrt{m})|j\rangle$. The displacement identity gives

$$\|(N/m - 1)\zeta_j\|^2 = (2j+1)/m + j^2/m^2.$$

Its coefficients are the Poisson amplitude times a fixed-degree Charlier polynomial. Explicitly, for the first three levels,

$$\begin{aligned}(\zeta_0)_n &= (-1)^n e^{-m/2} m^{n/2}/\sqrt{n!},\\ (\zeta_1)_n &= (m-n)(\zeta_0)_n/\sqrt{m},\\ (\zeta_2)_n &= ((n-m)^2 - n)(\zeta_0)_n/(\sqrt{2}m).\end{aligned}$$

For fixed j, the squared polynomial is bounded by $C_j m^j (1+n/m)^{2j}$ for $m \geq 1$.

Let $S_-|n\rangle = |n-1\rangle$ for $n \geq 1$ and $S_+ = S_-^\dagger$. Comparison to $a_1/\sqrt{m}$, followed by displacement, gives

$$\|(S_- + 1)\zeta_j\| \leq \|(N/m - 1)\zeta_j\| + \sqrt{j/m} \longrightarrow 0.$$

The analogous bound for S_+ includes the vanishing vacuum boundary term. Every fixed string of shifts therefore acts as its parity sign on ζ_j, up to norm error tending to zero.

On $m/2 \leq n \leq 2m$ and bounded k, the diagonal coefficients above converge uniformly to $d + k, 0, 0$. The E edge vanishes, while the $\widehat{B}$ and $\widehat{P}$ edges tend to $-\sqrt{(k+1)(2d+k)}/2$ and $-i\sqrt{(k+1)(2d+k)}/2$. The minus sign from S_- turns these into B_0, P_0. In (10), the common oscillator edges are $O_{d,k}(\sqrt{\epsilon m})$, while $s_k/\sqrt{\epsilon} \to \sqrt{(k+1)(2d+k)}$; hence $B_{\mathrm{tr}} \to -B_0$.

For a word of fixed length L, only $|\Delta n|, |\Delta k| \leq L$ occur. From $[n] \leq n$ and $1 - e^{-x} \geq x e^{-x}$, each coefficient has a bound of the form $C_{d,k,L}(1 + \epsilon(n+L))^L e^{L\epsilon(n+k+d+L)}$ after enlarging the constant. Squaring this bound and using the polynomial-times-Poisson estimate controls the omitted tails: exponential tilting changes the mean to $m e^{O_L(\epsilon)}$, and the prefactor is $e^{m(e^{O_L(\epsilon)}-1)} \to 1$. Outside $[m/2, 2m]$, exponential concentration beats every fixed polynomial. The finite path expansion of the word now proves (6). This argument controls products directly, rather than inferring them from convergence of factors.

Strong resolvents. Each finite-q LS combination is the self-adjoint direct sum of its finite total-level blocks; B_{tr} is bounded at fixed q. The limiting Jacobi operators are essentially self-adjoint on finite support since their off-diagonal coefficients grow linearly and satisfy the Carleman criterion. The same applies to $Q_0 = E_0 - P_0$; E_0 is diagonal. Core convergence implies strong-resolvent convergence for the conjugated individual operators, including $Q = E - \widehat{P}$. Indeed, for $w = (A_0 - z)v$, $v \in \mathcal{D}$, the resolvent difference is bounded by $|\operatorname{Im} z|^{-1}\|(A_0 - A_{q,m})v\|$; such w form a dense set. The uniform resolvent bound extends the result to the whole space. The corresponding unitary groups converge strongly at each fixed time.

5 A positive null-generator candidate

In the moving frame, define $Q_0 = E_0 - P_0$ and $T_0 = -B_0$, the limiting physical boost. Then

$$[T_0, Q_0] = iQ_0, \qquad Q_0 \geq 0. \tag{11}$$

The commutator follows from the displayed algebra. Positivity has a direct form proof: for finite-support $v = (v_k)$,

$$\langle v, Q_0 v\rangle = \frac{1}{2}\sum_{k\geq 0}\left|\sqrt{2d+k}\, v_k + i\sqrt{k+1}\, v_{k+1}\right|^2. \tag{12}$$

Its closure is nonnegative. A zero eigenvector would obey $v_{k+1} = i\sqrt{(2d+k)/(k+1)}v_k$, whose squared coefficients are proportional to $(2d)_k/k!$. They are not summable for $d > 0$, so zero is not an eigenvalue.

The phase transformation $v_k = i^k w_k$ turns Q_0 into the Laguerre Jacobi matrix with diagonal $d + k$ and off-diagonal $-\sqrt{(k+1)(2d+k)}/2$. The recurrence and orthogonality of $L_k^{(2d-1)}(2x)$ [4] give the seed spectral measure

$$d\nu_d(x) = \frac{2^{2d}}{\Gamma(2d)} x^{2d-1} e^{-2x}\, dx, \qquad x > 0. \tag{13}$$

Thus $\langle Q_0\rangle = d$, $\mathrm{Var}(Q_0) = d/2$, and $\langle e^{-aQ_0}\rangle = (2/(2+a))^{2d}$ for $a \geq 0$. This is the standard positive discrete-series realization already used as a kinematic model in the project's earlier translation note. The present step identifies its finite-word and resolvent approximation by actual chord generators in a specified moving frame; it does not create a new Laguerre representation.

In particular, Proposition 2 proves

$$\|([B_{\mathrm{tr}}, Q] - iQ)\mathcal{U}v\| \longrightarrow 0, \qquad v \in \mathcal{D}. \tag{14}$$

The finite-q matrix Q itself need not be positive. Equation (11) belongs to its self-adjoint limiting operator. The associated boost action rescales Q_0 as $e^{i\eta T_0} Q_0 e^{-i\eta T_0} = e^{-\eta} Q_0$.

The remaining translation problem. A single positive dilation-covariant operator is a null translation candidate. It is not a complete Poincare dictionary. The other LS combination $E_0 + P_0$ does not commute with Q_0, so the two cannot simply be called the two light-cone momenta. The earlier construction $P^+ = \mu Q_0$, $P^- = M^2/(\mu Q_0)$ still inserts M and μ by hand. Determining the opposite null action and its multiparticle composition from the microscopic model, rather than selecting an arbitrary inverse, is the next physical question. In particular $d = 2$ remains a representation weight, not a mass determination.

6 An insertion of the full selected density

The established covariant algebra has the self-adjoint fields h, U and the actual density

$$\rho = a + s(h - W)^2, \quad s = \sqrt{q}, \quad W = \sqrt{\lambda}U, \quad a = 1 - s(1 + \lambda) > 0. \tag{15}$$

Here U is a unit q-Gaussian. Its operator norm gives the uniform estimate

$$\|W\| \le \sqrt{\frac{\epsilon}{1 - e^{-\epsilon}}} = O(1). \tag{16}$$

This bound retains U on the full tracial Hilbert space; no energy-only replacement of ρ is made.

Specify the preparation being compared. Embed the one-matter module in the full tracial representation, choose a unit $v \in \mathcal{D}$, and let $\zeta = \mathcal{U}v$. Apply the density on the left:

$$\zeta_\rho = \frac{L_{\sqrt{\rho}}\zeta}{\|L_{\sqrt{\rho}}\zeta\|}. \tag{17}$$

This is a vector-insertion protocol. It is not automatically the original nonstationary functional $\tau(\rho X^\dagger \alpha_t(X))$, whose density ordering must be retained separately.

Proposition 3. *In the window of Proposition 2, set*

$$\eta = \frac{\|(h + 2\sqrt{m})\zeta\| + \|W\|}{2\sqrt{m}} + \frac{\sqrt{a}}{2\sqrt{sm}}.$$

Then

$$\left\| \left(\frac{\sqrt{\rho}}{2\sqrt{sm}} - 1 \right) \zeta \right\| \le \eta \longrightarrow 0, \tag{18}$$

$$\|\zeta_\rho - \zeta\| \le 2\eta. \tag{19}$$

For every bounded operator A in this common representation, the expectation difference is at most $4\eta\|A\|$. The squared insertion norm $\|L_{\sqrt{\rho}}\zeta\|^2$ divided by $4sm$ tends to one.

Proof. On the first factor $h = a_q + a_q^\dagger$. The inequality $0 \le 1 - \sqrt{[n]/n} \le \epsilon n/2$ bounds the q-oscillator error on ζ_j by $O_j(\epsilon m^{3/2})$. The ordinary displaced Hamiltonian has residual norm $\sqrt{2j+1}$ about $-2\sqrt{m}$. Hence $\|(h + 2\sqrt{m})\zeta\|/\sqrt{m} \to 0$ for finite v. Put $A_* = h - W$. Spectral calculus gives $\|(|A_*| - 2\sqrt{m})\zeta\| \le \|(A_* + 2\sqrt{m})\zeta\|$ and $\|\sqrt{a + sA_*^2} - \sqrt{s}|A_*|\| \le \sqrt{a}$. Together with (16) these prove the first bound. Normalizing a vector at distance at most η from a unit vector changes that distance to at most 2η. The bounded-observable estimate follows by two applications of Cauchy–Schwarz. □

For the true endpoint heat vector, the preceding stage supplies the same Hamiltonian concentration under the stronger window $\epsilon m^2 \to 0$, so this bound also applies there. It proves that this thermal insertion loses leading preparation information. Recovering the original selected response requires its ordering and subleading terms, not just this scalarized limit.

7 Verification and next calculation

The exact checks use $q = 1/4, 4/9$, $d = 1, 2, 3$ and total levels zero through seven: 1,224 Gram equalities and 864 ladder actions compare (1) with the original nonorthogonal LS representation. Twenty-one symbolic Laguerre recurrences and three exact finite positivity forms check the spectral step.

For moving vectors we use $m = 8, 32, 128, 512$, $\epsilon = m^{-3}$, $d = 2$, $j = 0, 1, 2$ and $k = 0, 1, 2, 3$ (48 cases). Each case tests nine words, all three commutators, the Casimir, and the boost match. The grid is padded beyond every applied shift. Two common-oscillator cutoffs differ by 48. The largest word refinement difference is 1.08×10^{-23}; a Poisson polynomial envelope bounds the omitted input norm squared by 3.18×10^{-48}. These are distinct checks: the input tail alone does not certify every unbounded word. Its analytic control is in the proof.

	$\|([\widehat{B}, \widehat{P}] - iE)\mathcal{U}\|j,k\rangle\|$		$\|([B_{\mathrm{tr}}, Q] - iQ)\mathcal{U}\|j,k\rangle\|$	
m	$(0,0)$	$(2,3)$	$(0,0)$	$(2,3)$
8	0.479014	3.206959	0.337907	2.461123
32	0.123308	1.043909	0.114994	0.933263
128	0.031132	0.266054	0.047929	0.374348
512	0.007805	0.066461	0.022581	0.172697

The errors on excited states are visibly larger than on the seed; a seed check would not establish the theorem's scope. Eight words on the largest-displacement $j = 2, k = 3$ vector were recomputed by an independent sparse direct-coefficient routine at 50 decimal digits. The largest relative vector difference is 1.40×10^{-13}. The $[\widehat{B}, \widehat{P}]$ cancellation ratio reaches 733.45, and the null-commutator ratio reaches 376.62. They are retained in the output. Floating-point diagnostics are not interval certificates. The other two commutators have cancellation ratios up to 4.03×10^8 and 1.52×10^9; their tiny residuals are precision-sensitive and are not covered by the eight-word high-precision comparison. Their convergence rests on Proposition 2. No successful earlier scientific driver was rerun or its output modified.

The next calculation should use the established multiparticle chord coproduct to determine whether the positive candidate Q_0 has the required additive action and what fixes its opposite null partner. In parallel, the insertion ordering in the actual selected correlator must be carried through this frame; its U contribution cannot be recovered from the leading scalarization alone. Four-bilocal and finite-G dynamics remain open. This note advances the microscopic generator map rather than declaring those problems solved.

References

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[4] NIST DLMF, 18.3 and 18.9, Laguerre orthogonality and recurrence.

[5] NIST DLMF, 18.28(iii), Al-Salam–Chihara polynomials.